SECTION–I(i) : (Maximum Marks : 45)

- This section contains **15** questions.
- Each question has **FOUR** options (A), (B), (C) and (D). **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If only the bubble corresponding to the correct option is darkened.

Zero Marks : 0 If none of the bubbles is darkened.

Negative Marks : –1 In all other cases

1. Which of the following process does not remove permanent hardness :

(A) Permutit process (B) Synthetic resins method

(C) Clark's method (D) Calgon's method

2. The radius of La^{+3} is 1.06 Å which of the following given values will be closest to the radius of Lu^{+3} (At. number of Lu = 71, La = 57)

(A) 1.6 Å (B) 1.4 Å (C) 1.06Å (D) 0.85 Å

3. Which of the following does not evolve any gaseous product on heating :

(A) $NaNO_3$ (B) Na_2CO_3 (C) NH_4NO_3 (D) $(NH_4)_2CO_3$

4. Which of the following compounds are coloured in aq. solution

(A) $Ce(SO_4)_2$ (B) $TiCl_4$ (C) $K_3[Cu(CN)_4]$ (D) $ZnSO_4.7H_2O$

5. Which of the following will act as best oxidizing agent.

(A) CrO_3 (B) MoO_3 (C) WO_3 (D) All of these

6. Which of the following process is used to prepare H_2O_2 by :-

(A) Oxidation of 2-ethylanthraquinon by O_2

(B) Oxidation of 2-ethylanthraquinol by O_2

(C) Reaction of 2-ethylanthraquinol by H_2O

(D) Reduction of 2-ethylanthraquinol by H_2

7. In a sample of temporary hard water which of the following water softening process does **NOT** produced any precipitate of cation responsible for hardness?

(A) Boiling of water

(B) Addition of lime water

(C) Addition of sodium hexameta phosphate

(D) Addition of sodium carbonate

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8. $[X] \underset{H^+}{\overset{OH^-}{\rightleftharpoons}} [Y]$

Both [X] and [Y] produce ink blue solution with H_2O_2 in presence of organic solvents and H_2SO_4, [X] & [Y] respectively

(A) MnO_4^-, MnO_4^{-2} (B) $Cr_2O_7^{-2}, CrO_4^{-2}$ (C) $CrO_4^{-2}, Cr_2O_7^{-2}$ (D) SO_3^{-2}, SO_4^{-2}

9. A colourless solid (X) on heating evolve CO_2 & gives white residue, which is soluble in water. Residue also give CO_2 on treatment with dil HCl.

The compound (X) is/are

(A) $LiHCO_3$ (B) $NaHCO_3$ (C) $CaCO_3$ (D) $Ca(HCO_3)_2$

10. Which of the following statement is **INCORRECT** for K_2MnO_4 ?

(A) It is prepared by MnO_2 in acidic medium

(B) It produce brown coloured compound on addition of water

(C) It produce $KMnO_4$ on addition of water

(D) It produce purple colour compound with O_3

11. Which of the following is not correct about the chemistry of 3d and 4f series elements?

(A) 3d elements show more oxidation state than 4f series elements

(B) The energy difference between 3d and 4s orbital is very little.

(C) Europium (II) is more stable than cerium (II)

(D) The magnetic character in 3d series elements increases from scandium to copper

12. In which of the following reaction, H_2O_2 is acting as a reducing agent?

(A) $SO_2 + H_2O_2 \longrightarrow H_2SO_4$

(B) $2KI + H_2O_2 \longrightarrow 2KOH + I_2$

(C) $Ag_2O + H_2O_2 \longrightarrow 2Ag + H_2O + O_2$

(D) $PbS + 4H_2O_2 \longrightarrow PbSO_4 + 4H_2O$

13. The lowest degree of paramagnetism is shown by :

(A) $MnSO_4.4H_2O$ (B) $FeSO_4.6H_2O$ (C) $CuSO_4.5H_2O$ (D) $NiSO_4.6H_2O$

14. When chromite ore is heated with Na_2CO_3 strongly in air followed by cooling and washing with water, then brown residue is obtained which is of

(A) Fe_2O_3 (B) $Fe_2O_3.FeO$ (C) Na_2CrO_4 (D) $FeO.Cr_2O_3$

15. In portland cement, which of the following compounds are mainly present. (I) Dicalcium silicate. (II) Tricalcium silicate (III) Tricalcium aluminate

(A) (I) only (B) (II) and (III) only (C) (I) and (III) only (D) I, II and III

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SECTION-I(ii) : (Maximum Marks: 12)

- This section contains **THREE** questions.
- Each question has **FOUR** options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s).
- For each question, choose the correct option(s) to answer the question.
- Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If only (all) the correct option(s) is (are) chosen.

Partial Marks : +3 If all the four options are correct but ONLY three options are chosen.

Partial Marks : +2 If three or more options are correct but ONLY two options are chosen, both of which are correct options.

Partial Marks : +1 If two or more options are correct but ONLY one option is chosen and it is a correct option.

Zero Marks : 0 If none of the options is chosen (i.e. the question is unanswered).

Negative Marks : –2 In all other cases.

16. When H_2O_2 reacts with $Ce(SO_4)_2$:-

(A) Ce^{+4} is oxidised to higher oxidation state

(B) Ce^{+3} is produced

(C) O_2 is produced

(D) H_2 is evolved

17. HD gas is prepared by :-

(A) Reaction of D_2O with NaH

(B) Reaction of H_2O with NaD

(C) Electrolysis of D_2O

(D) Na_2O_2 with D_2O

18. Reaction which involve redox change is/are

(A) $Cu^{+2} + \overline{C}N_{(Excess)} \longrightarrow$

(B) $FeS \xrightarrow{\text{Roasting}}$

(C) $CO_2 + MnO_4^- \xrightarrow{\text{weak alkaline solution}}$

(D) $Na_2CrO_4 + H_2SO_4 \longrightarrow$

SECTION–I(iii) : (Maximum Marks : 6)

- This section contains **ONE** paragraph.
- Based on each paragraph, there are **TWO** questions.
- Each question has **FOUR** options (A), (B), (C) and (D) **ONLY ONE** of these four options is correct.
- For each question, darken the bubble corresponding to the correct option in the ORS.
- For each question, marks will be awarded in one of the following categories :

Full Marks : +3 If only the bubble corresponding to the correct answer is darkened.

Zero Marks : 0 In all other cases.

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Paragraph Q. No. 19 to 20

Fajan's rule is very important for qualitative analysis of covalent character in ionic bond. According to this rule as the polarising power of cation and polarisibility of anion increases then covalent character in ionic bond increases.

Basis of Fajan's rule is also useful to predict various properties of ionic salt like nature of oxide, melting point, thermal stability and colour of ionic salt etc.

19. Select **INCORRECT** order of thermal stability.

(A) $Li_3N > Na_3N > K_3N$ (B) $LiCl < NaCl < KCl$

(C) $BeSO_4 < MgSO_4 < CaSO_4$ (D) $GeCl_4 > SnCl_4 > PbCl_4$

20. Select the **CORRECT** order of basic strength.

(A) $CaO < ZnO$ (B) $ZnO < HgO$ (C) $PbO > Ag_2O$ (D) $Fe_2O_3 > FeO$

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